A Identifiers

MGI-style example

Two read names

L1C009	R099	00001907
L1C009	R100	00001908

Shared template

R changes

Suffix changes

Encoded fields

B Nucleotide sequences

Without reference / unmatched short read

A C G T A C G T A C — preceding ten bases

G — next

Range coder

Context guides coding of the next base

With a reference: accepted match

Reference	A	C	G	T	T	G	C	A	A	C	C	G	T	A
Read	A	C	G	T	T	A	C	A	A	C	C	G	T	A

matching segments

Position + differences

C Quality scores

Sampled quality characters

I	I	I	I	I	H
H	?	@	@	G	G

Context model

Arithmetic coder

model bytes included

BSC

Block sorting + coding

Select smaller estimated size

Lossless quality stream

Illustrative schemes; all selected input bytes are restored.